

the problem involves an acute triangle abc with angle b equal to 75 degrees. the orthocenter h and circumcircle Ω are considered. points h_a and h_c are defined as the second intersections of lines ah and ch with Ω respectively. a circle centered at b with radius bh intersects line bc at x opposite to c . lines xh_a and ch intersect at p , while lines xh_c and ah intersect at q . an n good triangle satisfies $cp/ch = \sqrt{n}$ and aq being a positive integer. an amazing n exists if additionally $h_c q$ is an integer.

the sequence of amazing odd integers n_i grows exponentially. the growth rate x is derived from the minimal solution of a pell type equation arising from the trigonometric relationships in the triangle involving 75 degrees. the key equation $cp/ch = \sqrt{n}$ leads to a relation involving $\cot 75$ degrees, which is $2 - \sqrt{3}$. solving the resulting diophantine conditions yields a recurrence relation with characteristic root $2 + \sqrt{3}$. squaring this root gives $7 + 4\sqrt{3}$, the base x for n_{2i} . the constant z is 1 as the leading term dominates. thus $x + 2016 z$ equals $2023 + 4\sqrt{3}$, giving $p = 2023$, $q = 4$, $r = 3$. the sum $p + q + r$ is 2030 .

the exponential growth of n_i stems from solutions to the pell equation $x^2 - 4x - 1 = 0$, characteristic of the 75 degree angle. the base x is $(2 + \sqrt{3})^2 = 7 + 4\sqrt{3}$. with $z = 1$, the final expression simplifies to $2023 + 4\sqrt{3}$, yielding the required sum.